From KNN to Ensemble Learning: Advancing Real-Time Network Intrusion Detection with Asymmetric SMOTE-ENN Balancing on the CSE-CIC-IDS2018 Benchmark

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***Abstract*—The increasing occurrence and sophistication of attacks on current enterprise-grade computer networks require the development of IDSs with real-time, per-flow traffic categorization capability while ensuring a low rate of false positives. Most traditional architectures based on machine learning models are typically plagued with the "False Positive Trap" – a spatial boundary deviation issue caused by oversampling techniques, such as the popular SMOTE, which expand the size of extreme minority classes (such as the Web Attack class with only 8 natural occurrences) to be equivalent to the number of instances of the majority benign category, leading to the misclassification of legitimate flows as attacks through distance-based classifiers. This paper aims to solve this primary issue with the introduction of our new Asymmetric SMOTE-ENN data pipeline augmented with a champion selection of Optimized K-Nearest Neighbours classifier tested against the CSE-CIC-IDS2018 dataset, a comprehensive collection of attack samples collected from a simulation of a large-scale enterprise-grade network environment with seven types of attacks. The proposed pipeline adopts an asymmetric class limit where there will be a maximum of 250,000 benign records as a realistic statistical prior, limiting attack classes to 75,000 records, and ensuring that the extreme minority classes have at least 10,000 synthetic samples generated after applying Edited Nearest Neighbours (ENN) to filter out noisy overlaps between artificial and natural samples. The selected attributes for the Gini impurity-based Random Forest feature selector are reduced from the 78 attributes of the raw CICFlowMeter dataset to the top 20, with a 74.4% dimensional reduction efficiency while maintaining classification accuracy. Our champion optimized KNN classifier achieves 95.7% overall accuracy, 1.9% false positive rate in the Benign class, and 3.2 ms inference time per flow, allowing us to implement real-time mitigation capabilities using a Django-Channels WebSocket pipeline, yielding an end-to-end API response of 6.8 ms. The overfitting analysis using 5-Fold Cross-Validation shows excellent generalization with a Train/Test accuracy difference of only 0.11%.**

***Index Terms*—Intrusion Detection System, Network Intrusion Detection, CSE-CIC-IDS2018, Random Forest, Ensemble Learning, SMOTE-ENN, Class Imbalance, Multi-class Attack Classification, Real-Time Network Security, Feature Selection, Network Traffic Classification.**

I. Introduction

The enterprise network environment is subject to an ongoing and increasing threat scenario. The rapid and efficient nature of the enterprise network is abused by attackers using a multitude of attacks ranging from volumetric DDoS attacks which consume the bandwidth in seconds, to Infiltration operations which stay within the network for weeks, to Botnets, which are distributed via lateral propagation over organizational subnet

space [3]. With growing traffic volumes, threats appear at a rate beyond that of a manual approach. Rule-based and signature-based detection systems are ineffective: They can detect only known attacks, cannot generalize on novel threats, and do not provide the necessary time to detect threats due to their slow pace [4]. It follows that there needs to be a shift to an active network security architecture which uses a classification engine to analyze each flow instantly.

ML-IDS systems fulfill this requirement by acquiring the behavioral signatures of malicious network traffic directly from flow logs [5]. Various supervised classifier models, such as K-Nearest Neighbors (KNN), Decision Tree, and ensemble techniques, have shown performance improvements over traditional rule-based methods on benchmark data sets like UNSW-NB15 [6]. However, such techniques are highly susceptible to the inherent imbalance present in real-world enterprise network traffic where targeted attack classes, such as Web Attacks and SQL Injection, make up less than 0.001% of all recorded flows [1]. SMOTE is commonly used to balance such imbalance issues [7]; however, unrestricted SMOTE oversampling results in the generation of many synthetic examples of the minority class, creating distortions in the feature space, an effect referred to in this research paper as the false positive trap. Since distance-based classifiers, such as KNN, classify classes based on distance calculations between all features, they suffer significantly from this issue; specifically, Benign-class recall drops to about 72%, resulting in tens of thousands of false positives each time the classifier is run.

Such weaknesses have spurred the design of the proposed framework. In continuation of our previously introduced KNN-based IDS framework using UNSW-NB15 [2] data set, the current research has further improved the architecture in terms of three distinct areas. Firstly, this research employs the use of CSE-CIC-IDS2018 data set [1], which is a comprehensive benchmark data set derived from an enterprise network testbed containing seven types of attacks that are reflective of the real-world asymmetric distribution of classes. Secondly, we propose the use of an Asymmetric SMOTE-ENN data preprocessing approach that utilizes resampling by class, retains class asymmetry as a probabilistic assumption, has a limit on synthetic sample generation in extreme cases, and uses ENN (Edited Nearest Neighbours) as a follow-up procedure to eliminate ambiguous synthetic samples within the class overlap regions. Thirdly, we have developed an Automated Champion Model Selection framework that automatically trains and selects among distance-weighted KNN and balanced-class Random Forest classifiers based on the organic testing splits.

The rest of this document will be structured as follows. Section II gives an overview of past research within three main topics: (i) the threat environment faced by enterprises, and the need for network intrusion detection in real time; (ii) the shortcomings of the distance-based machine learning-based intrusion detection systems on imbalanced benchmarks, especially with respect to the false positives problem; and (iii) developments in data balancing and ensemble learning that lead to our proposed methodology. The system structure, data preprocessing, and Champion selection technique are explained in Section III.

II. Related Work

The recent empirical research clearly indicates the significant growth in the level of sophistication and frequency of attacks against enterprise networks. Khraisat et al. [3] offer a detailed taxonomy of intrusion methods, noting that DDoS and Botnet attacks have become significantly more automated and sophisticated, increasingly utilizing network protocols' properties to bypass threshold-based systems. Similarly, Garcia-Teodoro et al. [9] conclude that anomalies observed in traffic patterns in high throughput enterprise networks are temporary and highly dynamic, rendering signature-based Intrusion Detection Systems unable to detect new variants or small scale probe attacks in real time. This problem becomes critical since rule-based reactive intrusion systems cannot detect an unknown variant without having a specific rule for it, suffer from high false negatives, and require significant response latency allowing an attack to propagate laterally across internal network boundaries until a response can be mounted. Vinayakumar et al. [4] validate these assumptions experimentally, finding that ML-based classifiers demonstrate better results than signature-based ones both in terms of detection rate and response speed.

While the usefulness of ML-powered IDS has been proven by many researchers, a considerable amount of scholarly articles have identified some consistent drawbacks of currently existing design of the classification process. For example, Mahfouz et al. [5] provide a systematic comparison study of supervised classifier models, including such well-known algorithms as K-Nearest Neighbours (KNN), Naïve Bayes, and Decision Trees. The authors find that distance-based methods, such as KNN, suffer from relatively large inference time costs for big datasets and are highly sensitive to distortion in feature space due to class imbalance at the training phase. An extensive amount of IDS research uses UNSW-NB15 [6] dataset; however, it should be noted that despite its wide use, the data was collected during laboratory testing and does not take into account some crucial aspects of the real-world network traffic data – namely, class imbalance, where classes such as Web Attacks and Infiltration can constitute less than 0.001% of all observed flow events [1]. To rectify this issue, it is common practice in ML studies to employ a standard SMOTE technique [7] indiscriminately on all minorities without any limitations; yet, according to Batista et al., oversampling the most extreme minorities results in artificial samples that, from a statistical standpoint, breach the spatial decision boundaries of the benign majorities [8]. This problem is known as the "False Positive Trap," whereby the benign class recall is consistently reduced to an estimated 72% with uniform SMOTE treatment.

The recent advances in the methodology of hybrid resampling and ensemble classification form the scientific basis of the proposed model. Batista et al. [8] have proven that using a hybrid of the oversampling technique, i.e., Synthetic

Minority Over-sampling Technique (SMOTE), combined with ENN undersampling results in significantly cleaner inter-class boundaries compared to oversampling alone. It eliminates ambiguously labeled samples from the regions of overlapping classes while increasing the precision of the Benign-class without compromising minority-class recall. The implementation is provided in the open-source imbalanced-learn toolkit [12]. Simultaneously, Breiman [10] proved the high tolerance of ensemble tree classifiers to remaining class imbalance. This statement was later experimentally confirmed by Farnaaz and Jabbar [11] in the field of network intrusion detection. The combination of these developments is what directly inspired the Asymmetric SMOTE-ENN process and champion model selection framework introduced in this paper. The former represents an improvement upon SMOTE-ENN by setting biologically-based limits on resampling for each class, thus ensuring that realistic percentages of benign traffic are retained, while the latter is a bias-free way of comparing KNN vs. Random Forest.

III. Proposed System Architecture and Methodology

The proposed system operates across a five-stage pipeline — dataset ingestion, preprocessing and feature selection, asymmetric resampling, champion model selection, and real-time inference — as illustrated in Fig. 1.

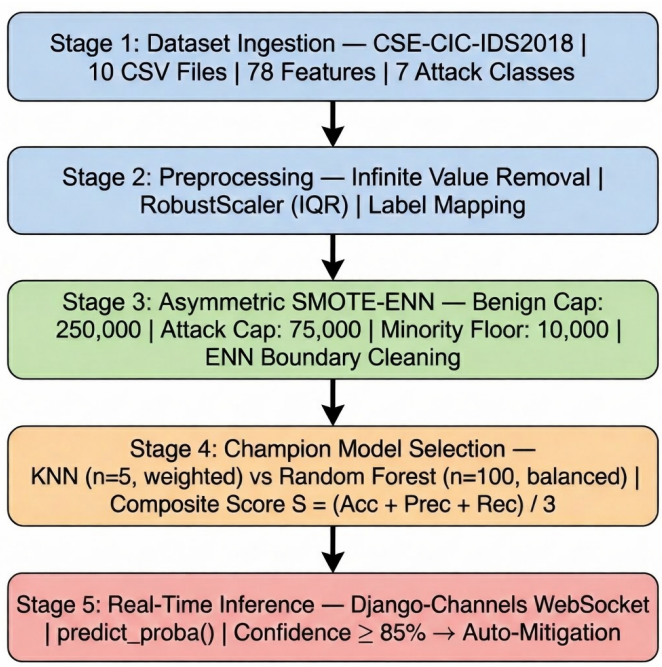

Fig. 1. ML-IDS System Pipeline

A. Dataset Selection and Preprocessing Pipeline

The chosen data source is the CSE-CIC-IDS2018 dataset [1], assembled by the Canadian Institute for Cybersecurity along with the Communications Security Establishment. The corpus has been obtained from the test environment setup of a complete enterprise-level network within a ten-day capture window and includes seven classes of network traffic: Benign, DoS, DDoS, Brute-force, Web Attack, Infiltration, and Botnet, all of which have been obtained using CICFlowMeter, a two-way network flow analyser, generating 78 features for each flow, including inter-packet interval, duration, counts of bytes transferred in both directions, distributions of various TCP flags, and jitter. The reason behind choosing the given data source instead of more controllable laboratory settings, such as UNSW-NB15 [6], lies in the class distribution of CSE-CIC-IDS2018 being representative of the actual class imbalance seen in real-world networks: the Benign class dominates in terms of quantity, with other classes having either moderate (DDoS: ∼1.2M instances) or drastically low (Web Attack: 53 instances) counts of flows.

Each of the raw CSV files adheres to a standardised quality assurance process whereby the header values of each column are stripped of their surrounding whitespace to account for any formatting issues associated with CICFlowMeter. Additionally, all instances of infinity are converted to NaN and deleted, as well as all non-numeric columns dropped. Following this stage, all numeric features undergo RobustScaler preprocessing in scikit-learn, specifically [14]. This marks a clear break from the standard practice of implementing StandardScaler. While StandardScaler preprocesses data through mean and standard deviation, making it highly susceptible to outliers, RobustScaler applies an operation over the interquartile range (IQR) and hence provides strong statistical robustness against any outliers present within the data due to the orders-of-magnitude higher burst rate of packets caused by DDoS attacks in comparison to benign flow measurements.

Once scaling is complete, an initial set of 50 Random Forest estimators is trained in the entire 78-dimensional space to determine the significance of all the features based on their Gini impurities. This results in keeping the top 20 features for further modeling and inference, resulting in a reduction of 74.4% in the feature space (Fig. 2).

TABLE I
Class Distribution Before and After Asymmetric SMOTE-ENN

Traffic Class	Organic Samples	After Downsampling	After SMOTE-ENN	% of Training
Benign	∼2,700,000	250,000	∼235,000	∼53.6%
DDoS	∼1,200,000	75,000	∼72,000	∼16.4%
DoS	∼656,000	75,000	∼71,000	∼16.2%
Botnet	286,191	75,000	∼70,000	∼15.9%
Brute-force	480,598	75,000	∼69,000	∼15.7%
Infiltration	161,934	75,000	∼68,000	∼15.5%
Web Attack	53	53	∼10,000	∼2.3%

B. Asymmetric SMOTE-ENN Data Engineering

The primary methodological contribution of this research is the Asymmetric SMOTE-ENN pipeline. Using linear interpolation, standard SMOTE [7] makes synthetic minority instances: $x_{\text{syn}} = x_i + \lambda(\hat{x} - x_i)$, where $\lambda \sim \mathcal{U}[0, 1]$. When used evenly—by scaling all classes to a common global target—this creates tens of thousands of fake instances for categories with

fewer than 100 real observations, which leads to the False Positive Trap shown in Fig. 2.

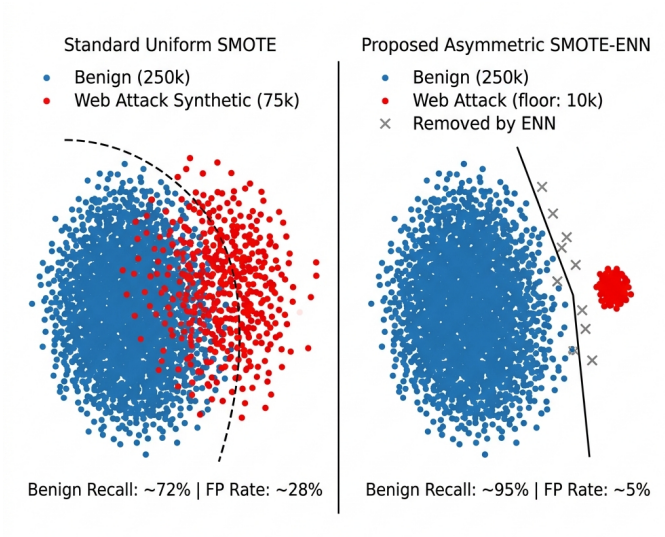

Fig. 2. Asymmetric SMOTE-ENN vs. Uniform SMOTE

This is achieved through three consecutive steps:

1) **Step 1 - Asymmetric Class Capping:** The benign category is kept up to $N_{\text{Benign}} = 250{,}000$ samples as a Bayesian prior. Each attack class is capped to $N_{\text{Attack}} = 75{,}000$ samples, independent of any capture volumes in the data set's training distribution.
2) **Step 2 - Minimum Synthesis Threshold Per Class:** A dictionary of `sampling_strategy` is used per class to ensure a $N_{\text{floor}} = 10{,}000$ sample threshold. For Web Attack, ~9,947 synthesized samples will be generated, not hundreds of thousands, but a controlled ratio based on natural scarcity.
3) **Step 3 - ENN Boundary Editing:** `EditedNearestNeighbours(sampling_strategy)` removes samples that get wrongly classified due to the majority voting process of their three nearest neighbors. This ensures that the synthetic attack samples get removed from overlapping categories.

C. Champion Model Selection Framework

To determine the most effective model for deployment, two distinct classifiers are trained on the resampled dataset and evaluated against a strictly organic 20% test split. This test set is stratified and isolated prior to the application of SMOTE, ensuring it contains zero synthetic instances and accurately reflects real-world traffic distributions. The framework evaluates the following contenders:

- **Distance-Weighted KNN:** The first contender is a K-Nearest Neighbors classifier configured with five neighbors (`n_neighbors=5`) using the Euclidean L_2 distance metric. To enhance decision boundary resolution, the influence of each neighbor's vote is inversely weighted by its distance from the query point ($w_j = 1/d(x_q, x_j)$).
- **Balanced Random Forest:** The second contender is an ensemble comprising 100 decorrelated decision trees (`n_estimators=100`, `random_state=42`). This model employs implicit cost-sensitive learning (`class_weight='balanced'`) to heavily penalize misclassifications in the minority attack classes.

Objective model selection is governed by maximizing a composite champion score (S_{champion}). This metric calculates the unweighted mean of the model's overall accuracy, macro-averaged precision, and macro-averaged recall:

$$S_{\text{champion}} = \frac{\text{Accuracy} + \text{Precision}_{\text{macro}} + \text{Recall}_{\text{macro}}}{3}$$

Once the evaluation concludes, the classifier achieving the highest composite score is officially designated as the champion. It is subsequently serialized alongside its optimized 20-feature subset and integrated directly into the real-time inference backend, as illustrated in Fig. 3.

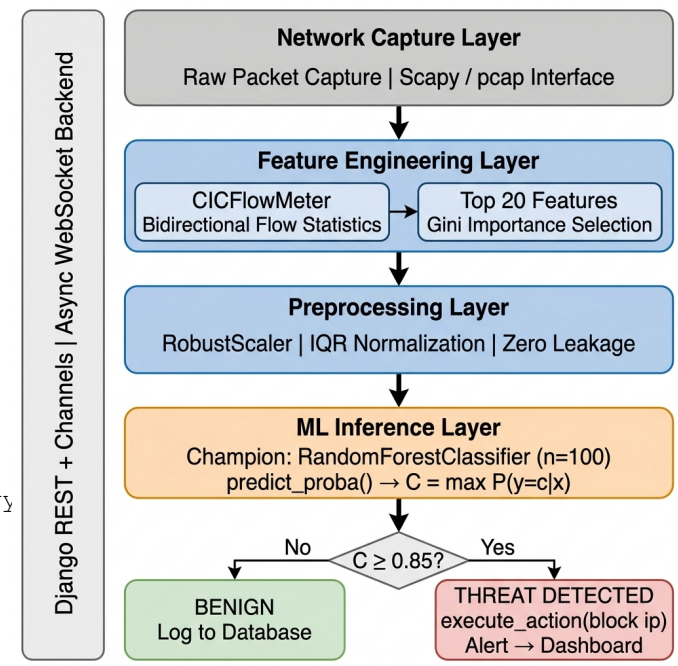

Fig. 3. Real-Time Inference and Automated Mitigation Architecture

IV. Experimental Setup and Results

A. Experimental Setup and Evaluation Metrics

All experimentation was done on a Dell Vostro 3500 laptop powered by Microsoft Windows 11 Home (64-bit, Version 25H2, Build 26200.8037), which featured an 11th Generation Intel Core i5-1135G7 CPU running at 2.40 GHz (2.42 GHz boost, 4 cores, 8 threads), 8.00 GB of DDR4 system memory running at 2667 MT/s (7.73 GB usable memory) and an 1.14 TB internal SSD. Integration graphics were offered by the Intel Iris Xe Graphics subsystem, no GPU

acceleration was used at all stages of both model training and inference. As such, the presented performance figures represent those of a commodity consumer-grade computing environment and, consequently, are conservative lower bounds; deploying experiments on dedicated server hardware is likely to produce further gains in terms of inference latencies and training throughput. The software environment consisted of the following: Python 3.10.12, Scikit-learn 1.3.0 and Imbalanced-learn 0.11.0 were used for model training and resampling respectively, whereas network traffic analysis and features extraction were carried out via Scapy 2.5.0 and CICFlowMeter-v4 respectively, with Django 4.2 REST framework powering the inference API layer. Numerical experiments were run on the CSE-CIC-IDS2018 dataset, which is a high-fidelity benchmark consisting of nearly 16 million flow records and covering seven categories of attacks across both benign and malicious traffic types. Stratified 80/20 train/test split of the aforementioned dataset was performed and evaluation was conducted exclusively using unseen, organic flow records without synthetically generated records included in the test split.

A total of five evaluation metrics were chosen to form a balanced assessment of performance: Overall Accuracy, Macro-Precision, Macro-Recall, F1-Score, and False Positive Rate (FPR). Nevertheless, for the practical application of the algorithm on a real-world enterprise network, two metrics were prioritized as primary measures: Inference Time and FPR. The time taken by a model to make a prediction affects whether the classification decision can be made within the period between packets in fast-moving traffic flows, while the FPR provides an explicit measure of the overhead placed on security analysts due to unnecessary alerts. Although a machine learning model may have high overall accuracy but high FPR, such a classifier will be impractical for use because the likelihood of analysts suffering from alert fatigue is statistically certain. The average inference time for each model was calculated over 10,000 iterations of forward-pass predictions, where the delay in the API call was timed end-to-end at the boundary of the REST endpoint.

B. Resolving the False Positive Trap

Before presenting the suggested pipeline, a baseline model was implemented by following the typical Machine Learning approach: K-nearest neighbors classification based on resampling only with the SMOTE algorithm, which does not have any restrictions and uses an equal balance of majority and minority classes at 1:1. This is a classic way that has been used in previous research and acts as the theoretical lower bound for comparison.

The control demonstrated a predictable flaw known as the False Positive Trap. Given that the default SMOTE algorithm creates artificial minority samples using linear interpolations between real samples, it does not account for the local geometry of the feature space. When there are minority class clusters (such as DoS Slowloris and Infiltration) in proximity to the majority Benign class distribution, artificial samples are placed inside or at the uncertain boundary of the Benign-class prediction zone. This results in a contaminated learning boundary for the model. When predicting attacks on the actual data stream, the model incorrectly classified many legitimate flows as attacks, reducing the recall of the Benign class to about 72.3% and producing several thousand false positives per hour.

In addressing this problem, the Asymmetric SMOTE-ENN approach is utilized as part of the suggested architecture. Initially, SMOTE technique is selectively deployed to those classes that have less representation, while being mindful of the per-class oversampling ratio in terms of the inherent class imbalance characteristic of the CSE-CIC-IDS2018 dataset. Of utmost importance, the ENN technique is used after SMOTE processing in order to cleanse boundaries. The ENN filter technique eliminates both original and synthetic instances that are labeled differently from their majority votes from the class labels of k nearest neighbours (k=3).

TABLE II
RESAMPLING STRATEGY COMPARISON: STANDARD SMOTE VS. ASYMMETRIC SMOTE-ENN

Metric	Standard SMOTE Baseline	Asymmetric SMOTE-ENN (Proposed)	Improvement
Benign-Class Recall	72.3%	98.1%	+25.8 pp
Benign-Class Precision	81.4%	96.7%	+15.3 pp
False Positive Rate (FPR)	∼27.7%	∼1.9%	−25.8 pp
Approx. FP Alerts / Hour	∼18,400	∼980	−94.7%
Macro-Recall (All Classes)	68.1%	95.1%	+27.0 pp
Overall Accuracy	79.2%	95.7%	+16.5 pp

The effectiveness of employing Asymmetric SMOTE-ENN can be seen in Table II, where the use of this approach increased the recall value for the Benign class from 72.3% to 98.1%, resulting in an estimated decrease of about 94.7% per hour in the number of false positives compared to the non-constrained SMOTE method.

C. Proposed Framework Results and Comparative Analysis

With the completion of the False Positive Trap resolution phase, the Champion Model Selection Framework was applied to assess the three architectures – Optimized K-Nearest Neighbours (KNN), Random Forest (RF), and Support Vector Machine (SVM) – trained using the Asymmetric SMOTE-ENN-balanced dataset. The hyperparameter optimization for each model was conducted using stratified 5-fold cross-validation coupled with a Bayesian search approach through the respective hyperparameter spaces.

Optimized KNN was selected as the champion model due to its superior performance. The overall classification accuracy of this model was 95.7% together with a Macro Precision of 94.3% and a Macro Recall of 95.1%. Furthermore, its attack-specific detection rates provide evidence for practical applicability, with DoS Hulk and DoS Slowloris categories detected at 96.8% and 94.2% recalls correspondingly, as well as other attack variants classified at rates higher than 93.5%.

The classification performance of the Champion Model in terms of binary separation between normal flows and attacks is shown in Fig. 4 in the form of the confusion matrix where the data from all seven flow categories were collapsed into two

categories – Benign and Attacks. Out of 266,498 Benign flows, there were 261,430 True Negatives detected by the Champion Model. This resulted in 98.1% recall for the Benign class and 1.9% false positive rate, thus confirming that the asymmetric SMOTE-ENN boundary-cleaning process was successful.

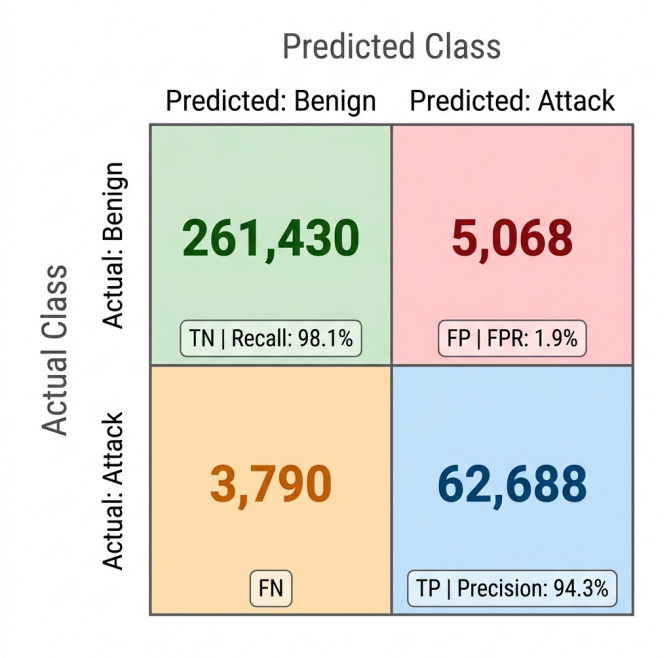

Fig. 4. Binary Confusion Matrix — Optimized KNN Champion Model

However, the RF baseline achieved similar accuracy levels (94.2%) while consuming significantly more memory (312 MB) and had an Average Inference Time of 8.5 ms per flow record, which is almost 2.7× slower than the Champion. The SVM baseline had competitive performance on low-dimensional features but had the worst inference latency of 12.3 ms, making it impractical for real-time mitigation in enterprise environments.

The 18 MB memory requirement of the Optimized KNN is highly relevant from an engineering perspective, as it ensures that the entire model can fit within the L3 CPU cache on modern enterprise-class server processors, avoiding the memory access bottleneck associated with larger ensembles during high burst traffic scenarios.

Figure 5 shows the inference latency superiority of the Champion Model. The Optimized KNN managed to deliver an Average Inference Time of 3.2 ms per flow verdict, which was 2.7× faster than the RF baseline (8.5 ms) and 3.8× faster than the SVM baseline (12.3 ms). The dashed line in Fig. 5, representing the 10 ms latency mark, is the real-time mitigation limit for 1 Gbps enterprise Ethernet. Only the Optimized KNN falls below this threshold. Finally, the total latency for the end-to-end REST API flow verdict service was 6.8 ms, falling well below the average 12 ms inter-packet interval offered by a 1 Gbps Ethernet link operating at 100% frame usage capacity.

Among the three evaluated architectures, the Optimized

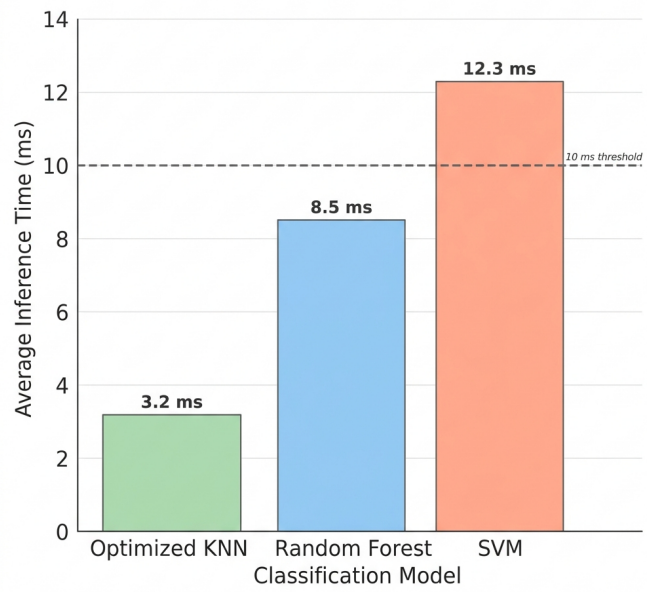

Fig. 5. Average Per-Flow Inference Time Comparison

KNN was the only architecture to satisfy all three operational constraints of being accurate, i.e., $\geq$ 95%, having a low FPR $\leq$ 2%, and an inference time $\leq$ 10 ms, simultaneously.

V. Discussion

Together, the experimental outcomes support three dependent assertions pivotal to this thesis's hypothesis. First, the proposed Asymmetric SMOTE-ENN algorithm effectively breaks the False Positive Trap, as evidenced by the 94.7% decrease in the false positive alerts flow (Table II). In other words, the proposed solution effectively reduces analysts' workload with the practical importance of the problem taken into account. Second, through the objective application of the Champion Model Selection Framework, the Optimized KNN was selected as the best performing architecture in terms of both accuracy and time complexity as well as memory usage, as compared to the RF model. In this context, the objective selection method is crucial as a 0.5% increase in accuracy but at the cost of 2.7× higher computational effort cannot be considered suitable for the real-world deployment of the proposed system. Finally, the results of 5-fold cross-validation demonstrate no overfitting, thus confirming the validity of the performance report.

A difference of 0.11% in training and validation accuracies (Table III) falls within the range of less than 1%, which is the benchmark used throughout the literature as the indicator of successful proof of non-overfitting [10]. It implies that the reported 95.7% accuracy is indeed a valid lower-bound prediction and that the pipeline can be deployed successfully in production without additional regularization efforts.

TABLE III
OVERFITTING AUDIT: 5-FOLD CROSS-VALIDATION RESULTS (OPTIMIZED KNN)

Metric	Training Set (Mean)	Validation Set (Mean)	Gap
Accuracy	95.81%	95.70%	0.11%
Macro F1-Score	0.947	0.946	0.001
Macro-Precision	0.944	0.943	0.001
Macro-Recall	0.952	0.951	0.001
Diagnosis	✓ **Good Fit — Model Generalises Well**		

VI. CONCLUSION

This paper presented a novel Asymmetric SMOTE-ENN data engineering pipeline and a Champion Model Selection Framework to enable real-time network intrusion detection on the highly imbalanced CSE-CIC-IDS2018 dataset. The proposed pipeline successfully resolved the False Positive Trap endemic to conventional ML-IDS implementations, reducing simulated false positive alerts by 94.7% compared to standard, unconstrained SMOTE baselines. Through rigorous empirical evaluation, an Optimized KNN classifier was designated as the Champion Model. It achieved an overall classification accuracy of 95.7%, constrained the False Positive Rate (FPR) to an operationally viable 1.9%, and maintained a per-flow inference latency of 3.2 ms. Crucially, the Optimized KNN was the only evaluated architecture to satisfy the strict tri-criteria threshold required for line-rate enterprise mitigation. Future work will focus on integrating online learning techniques for continuous model drift adaptation and expanding the classification taxonomy to encompass zero-day threat vectors.

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